Assignment 1

1] Proof that Eqn. 1 is derivative of Eqn. 2

$$q(t) = \sum_{k=0}^{\infty} \frac{(\hat{\omega} t)^k}{k!} q(0)$$

$$= \left(1 + \hat{\omega} t + \frac{(\hat{\omega} t)^2}{2!} + \dots + \frac{(\hat{\omega} t)^k}{k!} + \dots \right) q(0)$$

$$\dot{q}(t) = \left(0 + \hat{\omega} + \frac{2\hat{\omega}^2 t}{2!} + \dots + \frac{k \hat{\omega}^k t^{k-1}}{k!} + \dots \right) q(0) \qquad \| \frac{k}{k!} = \frac{1}{(k-1)!}$$

$$= \hat{\omega} \left(1 + \hat{\omega} t + \dots + \frac{(\hat{\omega} t)^{k-1}}{(k-1)!} + \dots \right) q(0) \qquad \| \text{Let } m = k-1$$

$$= \hat{\omega} \sum_{m=0}^{\infty} \frac{(\hat{\omega} t)^m}{m!} q(0) = \underline{\underline{\hat{\omega} q(t)}} \qquad \square$$

2] Show $\underline{\hat{\omega}^2 = \omega\omega^T - \|\omega\|^2 Id_3}$

$$\longrightarrow \hat{\omega}^2 = (1 - 1^T)^2$$

$$= \left(\begin{bmatrix} 0 & 0 & 0 \\ \omega_3 & 0 & 0 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} - \begin{bmatrix} 0 & \omega_3 & -\omega_2 \\ 0 & 0 & \omega_1 \\ 0 & 0 & 0 \end{bmatrix} \right)^2$$

$$= \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}$$

$$= \underline{\underline{\begin{bmatrix} -\omega_2^2 - \omega_3^2 & \omega_1\omega_2 & \omega_1\omega_3 \\ \omega_1\omega_2 & -\omega_1^2 - \omega_3^2 & \omega_2\omega_3 \\ \omega_1\omega_3 & \omega_2\omega_3 & -\omega_1^2 - \omega_2^2 \end{bmatrix}}}$$

$$\longrightarrow \omega\omega^T = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \cdot \begin{bmatrix} \omega_1 & \omega_2 & \omega_3 \end{bmatrix} = \underline{\begin{bmatrix} \omega_1^2 & \omega_1\omega_2 & \omega_1\omega_3 \\ \omega_1\omega_2 & \omega_2^2 & \omega_2\omega_3 \\ \omega_1\omega_3 & \omega_2\omega_3 & \omega_3^2 \end{bmatrix}}$$

$$\longrightarrow \|\omega\|^2 Id_3 = \omega_1^2 + \omega_2^2 + \omega_3^2 \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \underline{\begin{bmatrix} \omega_1^2 + \omega_2^2 + \omega_3^2 & 0 & 0 \\ 0 & \omega_1^2 + \omega_2^2 + \omega_3^2 & 0 \\ 0 & 0 & \omega_1^2 + \omega_2^2 + \omega_3^2 \end{bmatrix}}$$

$$\hookrightarrow \omega\omega^T - \|\omega\|^2 Id_3 = \begin{bmatrix} \omega_1^2 & \omega_1\omega_2 & \omega_1\omega_3 \\ \omega_1\omega_2 & \omega_2^2 & \omega_2\omega_3 \\ \omega_1\omega_3 & \omega_2\omega_3 & \omega_3^2 \end{bmatrix} - \begin{bmatrix} \omega_1^2+\omega_2^2+\omega_3^2 & 0 & 0 \\ 0 & \omega_1^2+\omega_2^2+\omega_3^2 & 0 \\ 0 & 0 & \omega_1^2+\omega_2^2+\omega_3^2 \end{bmatrix}$$

$$= \begin{bmatrix} -\omega_2^2-\omega_3^2 & \omega_1\omega_2 & \omega_1\omega_3 \\ \omega_1\omega_2 & -\omega_1^2-\omega_3^2 & \omega_2\omega_3 \\ \omega_1\omega_3 & \omega_2\omega_3 & -\omega_1^2-\omega_2^2 \end{bmatrix} = \hat{\omega}^2 \quad \square$$

Show $\hat{\omega}^3 = -\|\omega\|^2 \hat{\omega}$

$$\longrightarrow \hat{\omega}\omega = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \cdot \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \begin{bmatrix} \omega_2\omega_3 - \omega_2\omega_3 \\ \omega_1\omega_3 - \omega_1\omega_3 \\ \omega_1\omega_2 - \omega_1\omega_2 \end{bmatrix} = 0$$

$$\hookrightarrow \hat{\omega}^3 = \hat{\omega}\hat{\omega}^2 = \hat{\omega}(\omega\omega^T - \|\omega\|^2 Id_3)$$

$$= \hat{\omega}\omega\omega^T - \|\omega\|^2 \hat{\omega} Id_3 \qquad \| \ \hat{\omega}\omega = 0$$

$$= 0 - \|\omega\|^2 \hat{\omega} Id_3 \qquad \| \ \hat{\omega} Id_3 = \hat{\omega}$$

$$= -\|\omega\|^2 \hat{\omega} \quad \square$$

3] Show $R(\omega,\theta) = \exp(\hat{\omega}\theta) = Id_3 + \hat{\omega}\sin(\theta) + \hat{\omega}^2(1-\cos(\theta))$

$\longrightarrow$ Let $\|\omega\| = 1$. As $\hat{\omega}^3 = -\|\omega\|^2\hat{\omega} = -\hat{\omega}$, $\hat{\omega}^n$ follows the periodic pattern:

$\longrightarrow$ Odd case $(n = 2k+1)$: $\hat{\omega}^1 = \hat{\omega}$, $\hat{\omega}^3 = -\hat{\omega}$, $\hat{\omega}^5 = \hat{\omega}$, $\ldots \Rightarrow \hat{\omega}^{2k+1} = (-1)^k \hat{\omega}$

$\longrightarrow$ Even case $(n = 2k)$: $\hat{\omega}^2 = \hat{\omega}^2$, $\hat{\omega}^4 = -\hat{\omega}^2$, $\hat{\omega}^6 = \hat{\omega}^2$, $\ldots \Rightarrow \hat{\omega}^{2k} = (-1)^{k-1}\hat{\omega}^2$

$$R(\omega,\theta) = \exp(\hat{\omega}\theta)$$

$$= \sum_{n=0}^{\infty} \frac{(\hat{\omega}\theta)^n}{n!}$$

$$= Id_3 + \sum_{n=1}^{\infty} \frac{\theta^n}{n!}\hat{\omega}^n \qquad \| \ \text{Split into odd and even terms}$$

$$= Id_3 + \sum_{k=0}^{\infty} \frac{\theta^{2k+1}}{(2k+1)!}(-1)^k\hat{\omega} + \sum_{k=1}^{\infty} \frac{\theta^{2k}}{(2k)!}(-1)^{k-1}\hat{\omega}^2$$

$$= Id_3 + \hat{\omega} \sum_{k=0}^{\infty} (-1)^k \frac{\theta^{2k+1}}{(2k+1)!} + \hat{\omega}^2 \left(\sum_{k=1}^{\infty} (-1)^{k-1} \frac{\theta^{2k}}{(2k)!} \right) \quad \| \text{Def } \sin, \quad \| (-1)^{k-1} = -(-1)^k$$

$$= Id_3 + \hat{\omega} \sin(\theta) + \hat{\omega}^2 \left(-\sum_{k=1}^{\infty} (-1)^k \frac{\theta^{2k}}{(2k)!} \right)$$

$$= Id_3 + \hat{\omega} \sin(\theta) + \hat{\omega}^2 \left((-1)^0 \frac{\theta^{2\cdot 0}}{(2\cdot 0)!} - \sum_{k=0}^{\infty} (-1)^k \frac{\theta^{2k}}{(2k)!} \right) \quad \| \text{Def } \cos$$

$$= Id_3 + \hat{\omega} \sin(\theta) + \hat{\omega}^2 \left(1 - \cos(\theta) \right) \qquad \square$$